

Centre College Sports Analytics

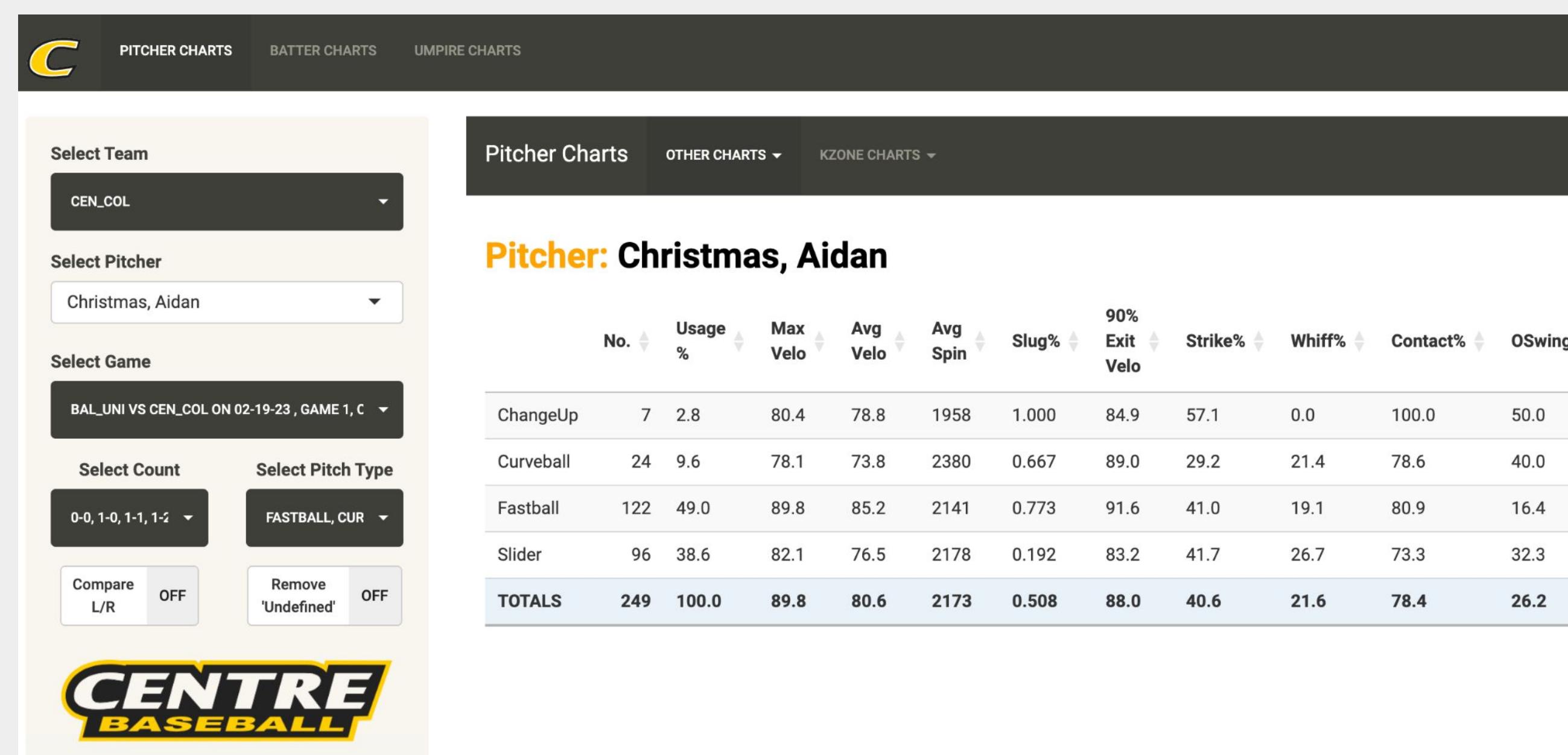
Gabriel Eze, Nolan Johnson, Colby Kinser, Jessie Strader, and Johnson Subedi
Centre College

Background:

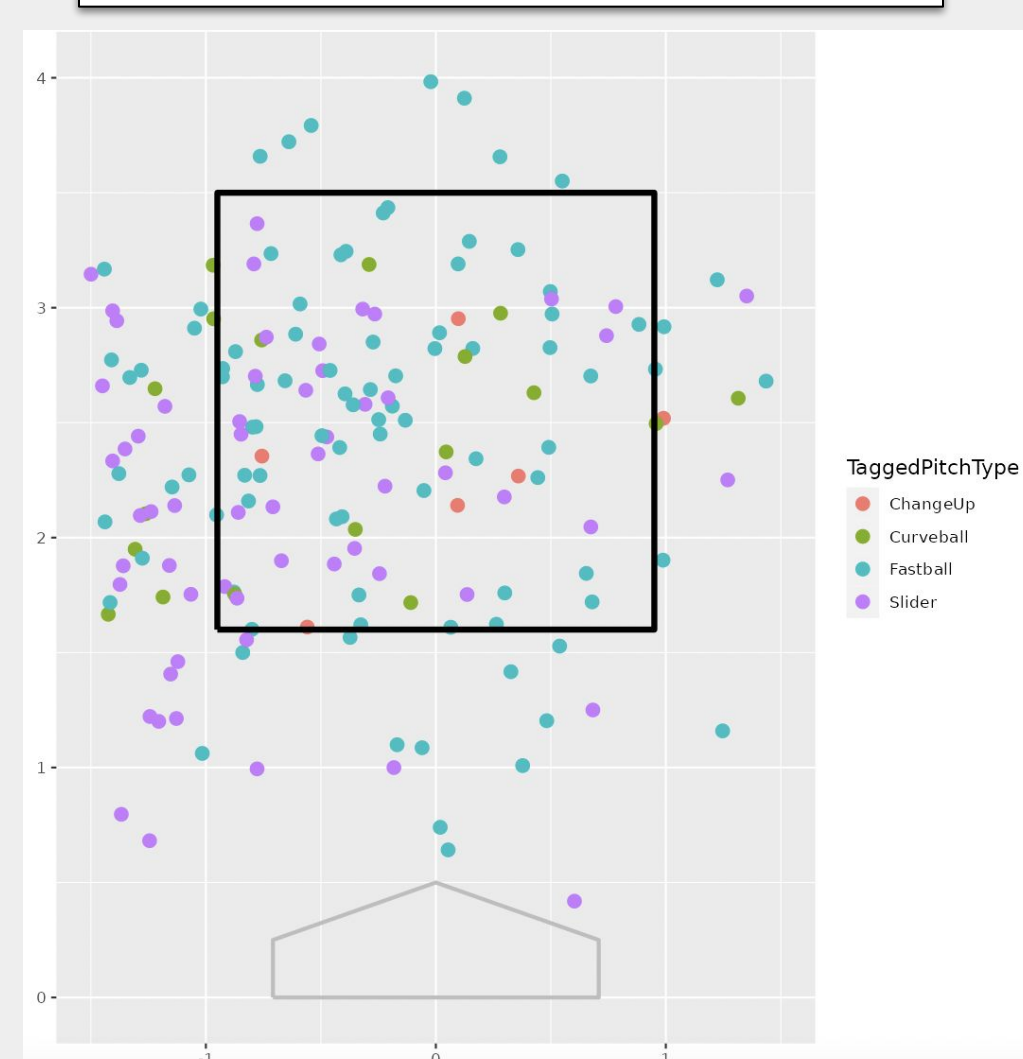
Centre College is a Division III school located in Danville, Kentucky. Centre is home to 25 different sports teams and nearly 50% of the school's approximately 1400 students are student athletes. Centre's sports analytics group is comprised of several faculty and students who are focused on using the data collected from Centre's athletics teams to help the teams succeed. The sports analytics group has worked with several of the school's teams to develop techniques and make data-driven insights which help these teams make better strategic decisions. Many of these analytics projects are centered around the development of apps using R Shiny which teams can then use to make informed decisions both during games and in their overall preparation. The group has begun to develop these apps for several teams including baseball, basketball, and volleyball.

Baseball

Overview: We use data from Trackman, a doppler radar flight tracking system, that measures a host of values on pitching (e.g., velocity, location, spin rate), hitting (e.g., exit velocity, launch rate), and fielding (e.g., player location). We developed an R Shiny app that computes various analytics and visualizations for coaches and players to easily access and filter.



Pitch Location by Pitch Type



Pitch Velocity by Pitch Count

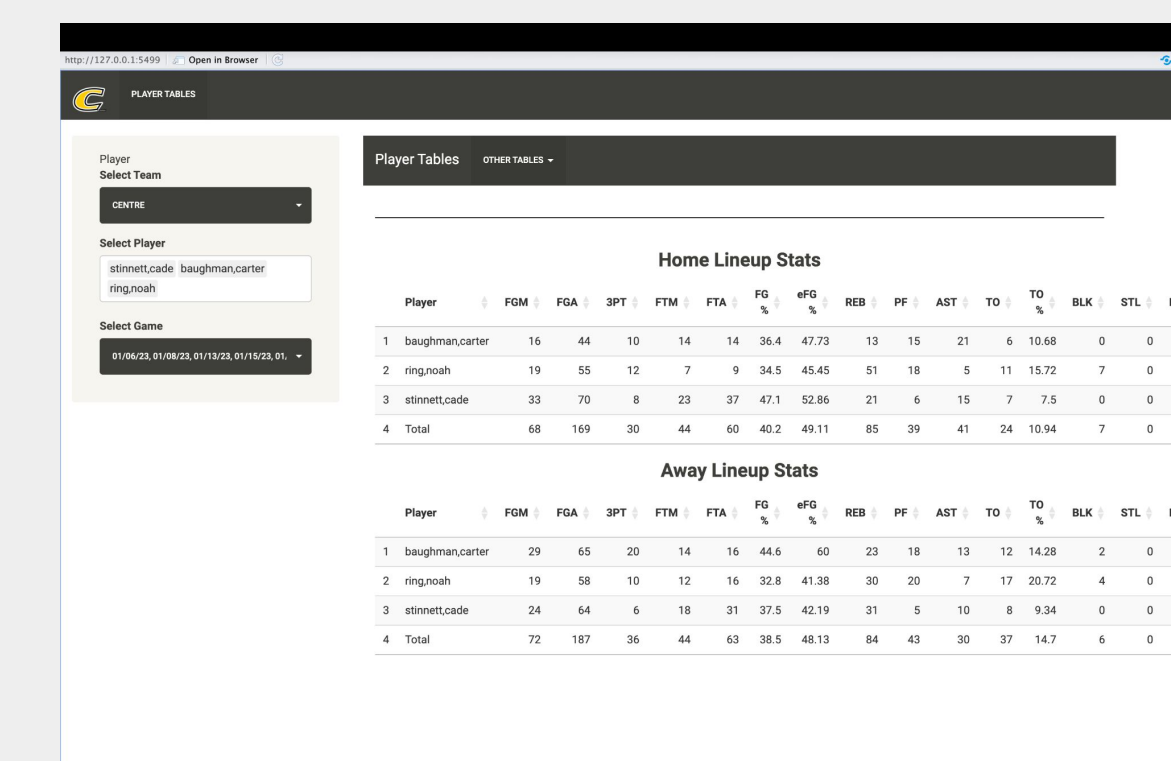
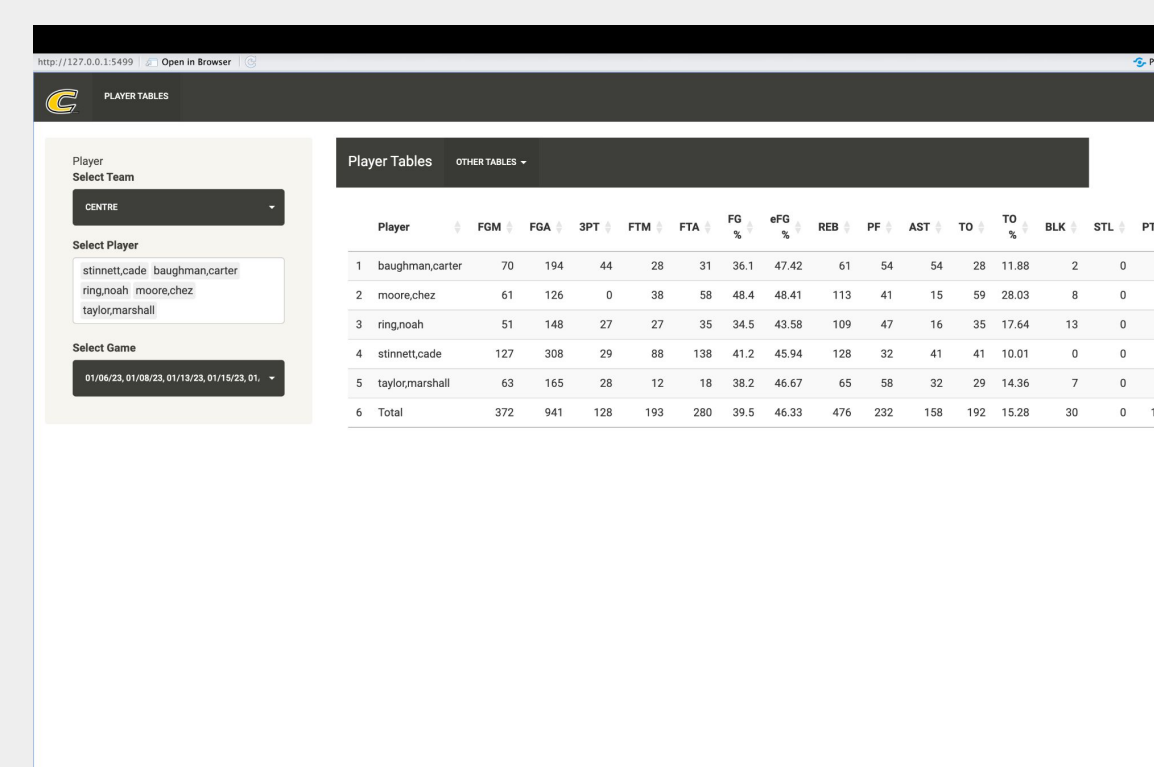


Pitch Classification: The Trackman has the ability to classify pitches, but given the classification is often inaccurate, we wanted to develop a more accurate automatic pitch classification system. We developed a multinomial logistic regression model to predict pitch types, and used a train/test method to test the accuracy of our model.

[Table Here:](#)

Basketball

Objective: Use data from schools in the Southern Athletic Conference (SAA) to calculate possession based statistics and conduct lineup analysis for Centre's basketball team.



Data Pipeline

Scrape Play-by-Play HTML from the Web using Python

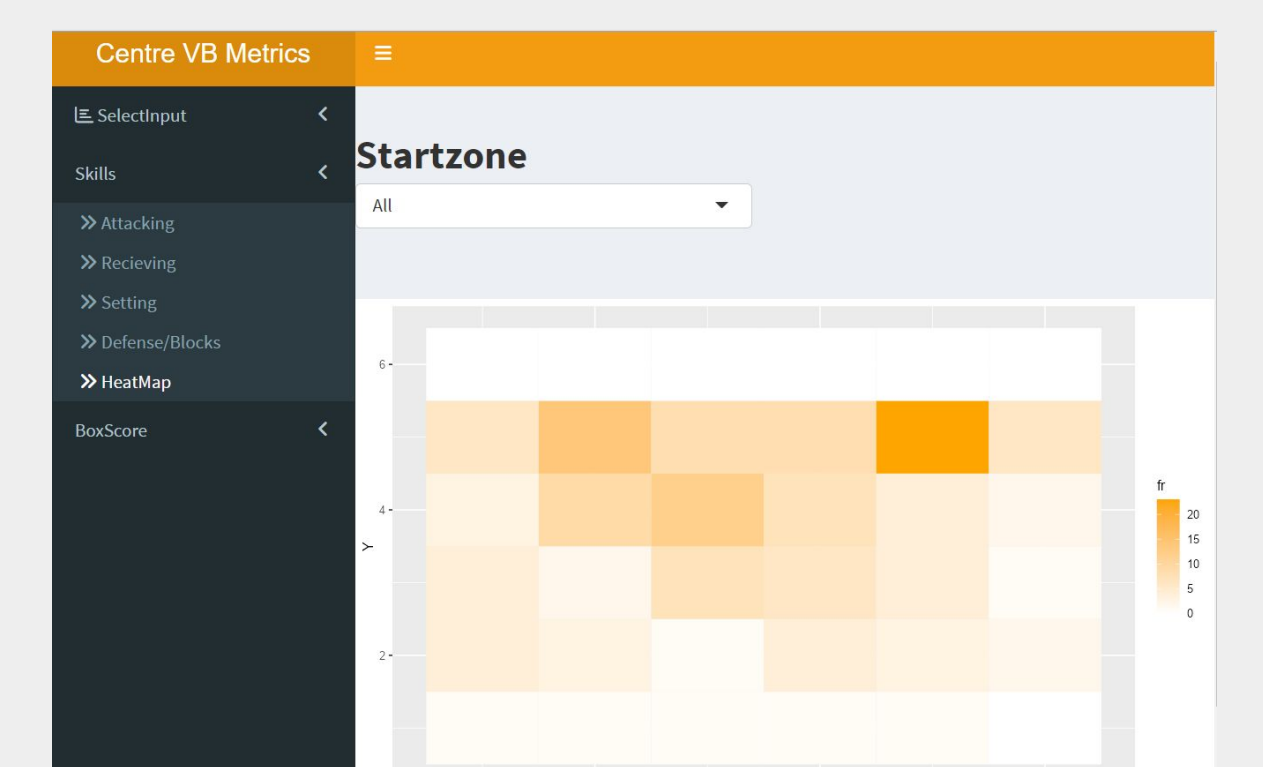
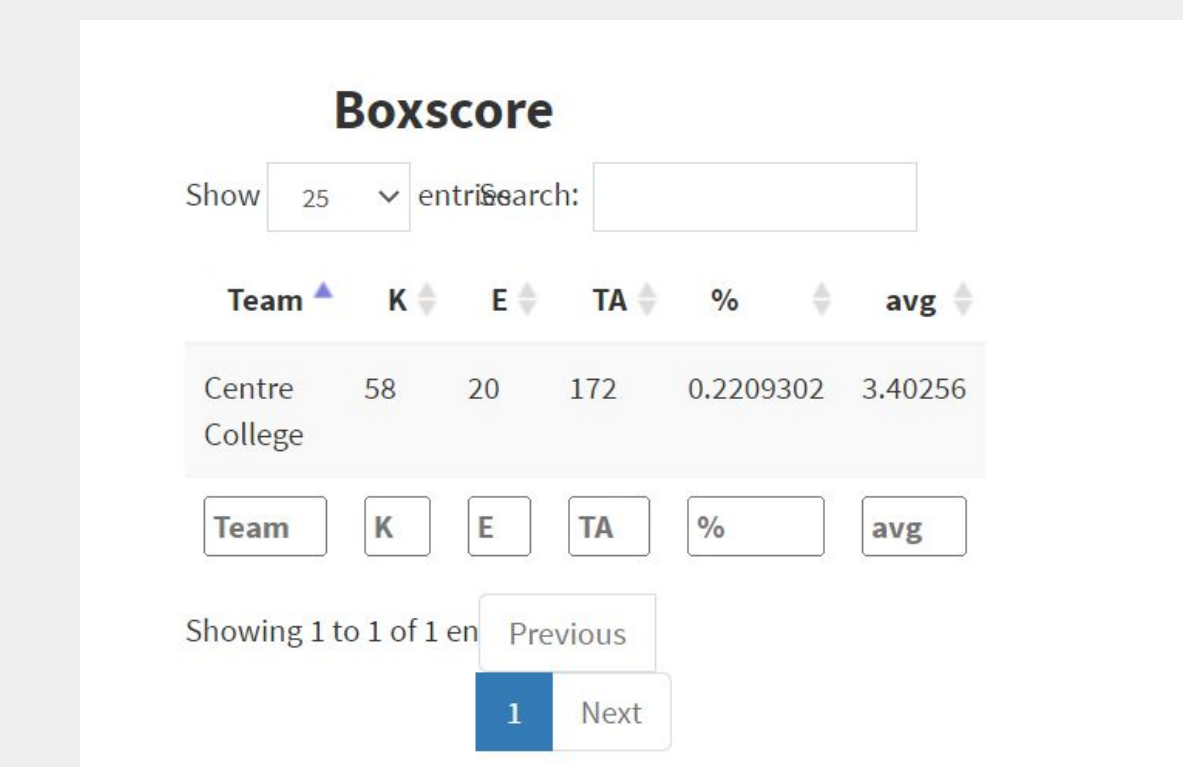
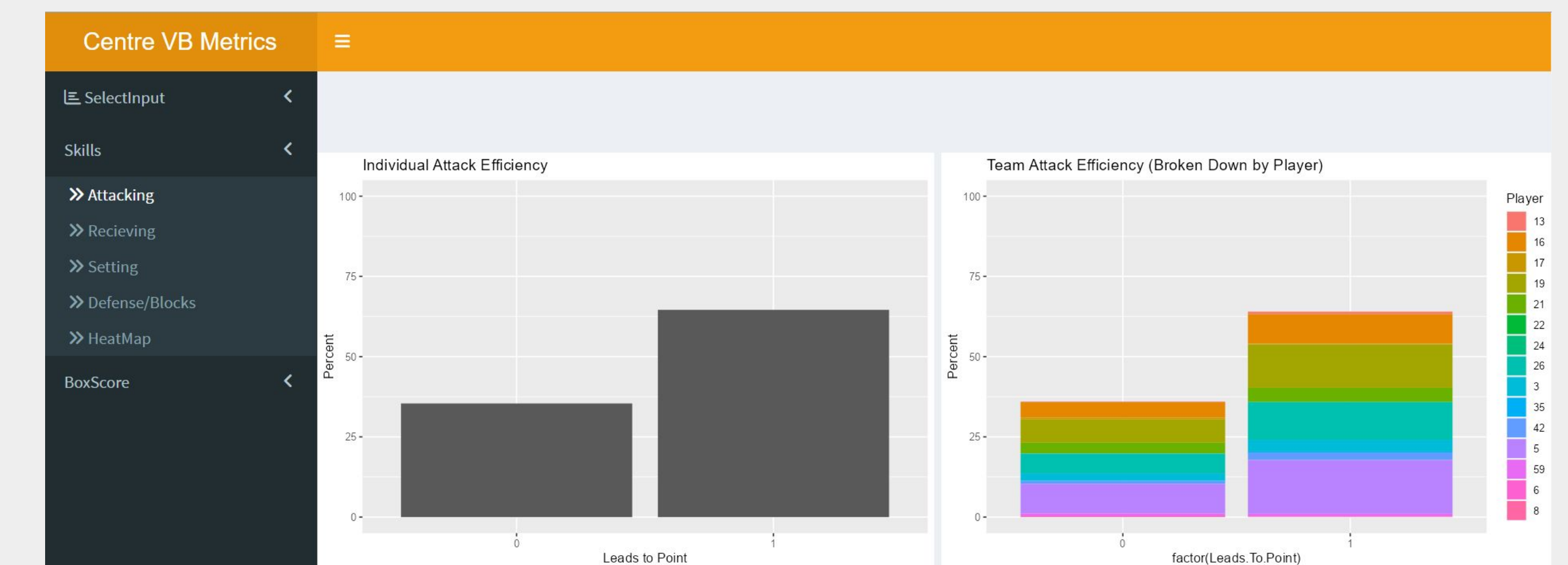
Process HTML Data into CSV, and Calculate Additional Fields

Compute Analytics in R

Upload to R Shiny App

- Data is available in a play-by-play format. Our processing calculates possession based statistics as well.

Volleyball



Data Pipeline

DataVolley returns DVW Encoded Match Data

Process DVW Data into CSV

Calculate Additional Fields

Compute Analytics in R

Upload to R Shiny App

- Centre submits match videos to third party provider, DataVolley.
- Data is returned to us encoded in a DVW file which we convert to game-by-game CSV files using Python.
- R is used to calculate standard analytics and player statistics such as attack efficiency and box scores.

References:

- Men's basketball cumulative statistics.* (n.d.). Centre College Athletics. Retrieved October 31, 2023, from <https://centrecolonels.com/sports/mens-basketball/stats/>

Acknowledgments: